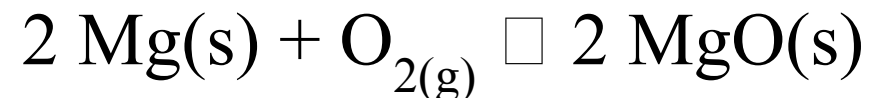
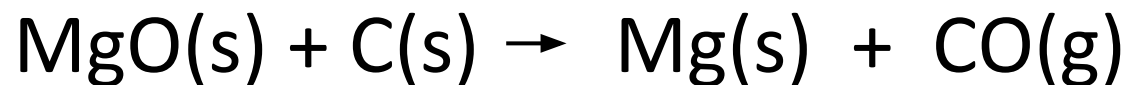


Oxidation-Reduction Reactions

- The term oxidation was originally used to describe reactions in which an element combines with oxygen.
- Example: The reaction between magnesium metal and oxygen to form magnesium oxide involves the oxidation of magnesium.



- The term reduction comes from the Latin stem meaning "to lead back." Anything that leads back to magnesium metal therefore involves reduction.
- The reaction between magnesium oxide and carbon to form magnesium metal and carbon monoxide is an example of the reduction of magnesium oxide to magnesium metal.



Oxidation–Reduction Reactions

Involves 2 Steps

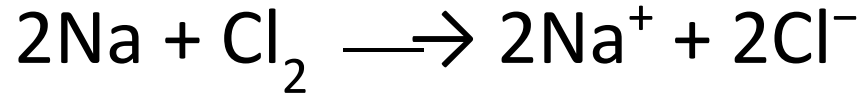
Oxidation = Loss of Electrons (LEO)



Reduction = Gain of electrons (GER)



Net reaction:



- Oxidation & reduction **always** occur together
- Can't have one without the other

Remember this phrase: LEO the lion says GER.

- **LEO** = Loss of Electrons is Oxidation
- **GER** = Gain of Electrons is Reduction
- Another way is to simply remember that reduction is to reduce the oxidation number.
- Therefore, oxidation must increase the value.
- **Oxidizing Agent** - that substance which oxidizes somebody else. It is reduced in the process.
- **Reducing Agent** - that substance which reduces somebody else. It is oxidized in the process.

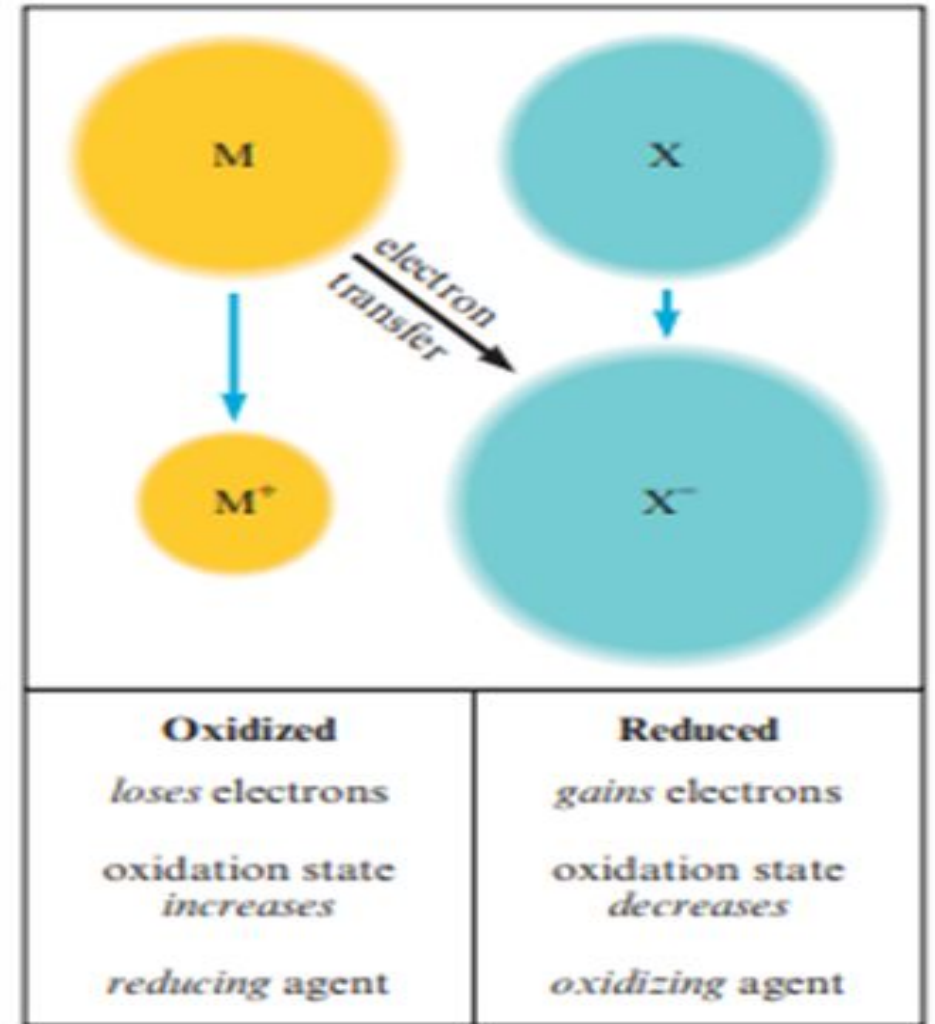


Figure 4.20 | A summary of an oxidation–reduction process, in which M is oxidized and X is reduced.

Predicting oxidation numbers

- Oxidation number of atoms in element is zero in all cases
- Oxidation number of element in monatomic ion is equal to the charge
- sum of the oxidation numbers in a compound is zero
- sum of oxidation numbers in polyatomic ion is equal to the charge
- F has oxidation number -1
- H has oxidn no. $+1$; except in metal hydrides where it is -1
- Oxygen is usually -2 . Except:
 - O is -1 in hydrogen peroxide, and other peroxides
 - O is $-1/2$ in superoxides KO_2
 - In OF_2 O is $+2$

Ex. Assigning Oxidation Number



$$\text{Li (2 atoms)} \times (+1) = +2$$

$$\text{O (1 atom)} \times (-2) = -2$$

$$\text{sum} = 0$$

$+2 - 2 = 0$ so the charges are balanced to zero



$$\text{C (1 atom)} \times (x) = x$$

$$\text{O (2 atoms)} \times (-2) = -4$$

$$\text{sum} = 0$$

$$x - 4 = 0 \text{ or } x = +4$$

C is in +4 oxidation state

Learning Check

Assign oxidation numbers to all atoms:

Ex. ClO_4^-

$$\text{O (4 atoms)} \times (-2) = -8$$

$$\text{Cl (1 atom)} \times (-1) = -1$$

(molecular ion) sum $\neq -1$

$$\text{O (4 atoms)} \times (-2) = -8$$

$$\text{Cl (1 atom)} \times (x) = x$$

$$\text{sum} = -1 - 8 + x = -1 \quad \text{or} \quad x = 8 - 1$$

So $x = +7$; Cl is oxidation state +7

Learning Check

Assign Oxidation States To All Atoms:



Mg = +2; O = -2; and Cr = x (unknown)

$$+2 + 2x + \{7 \times (-2)\} = 0$$

$$2x - 12 = 0 \quad x = +3$$

Cr is oxidation # of +3



K = +1; O = -2; so Mn = x

$$+1 + x + \{4 \times (-2)\} = 0$$

$$x - 7 = 0 \quad x = +7$$

Mn is oxidation # of +7

Redefine Oxidation-Reduction in Terms of Oxidation Number

- A redox reaction occurs when there is a change in oxidation number.

Oxidation

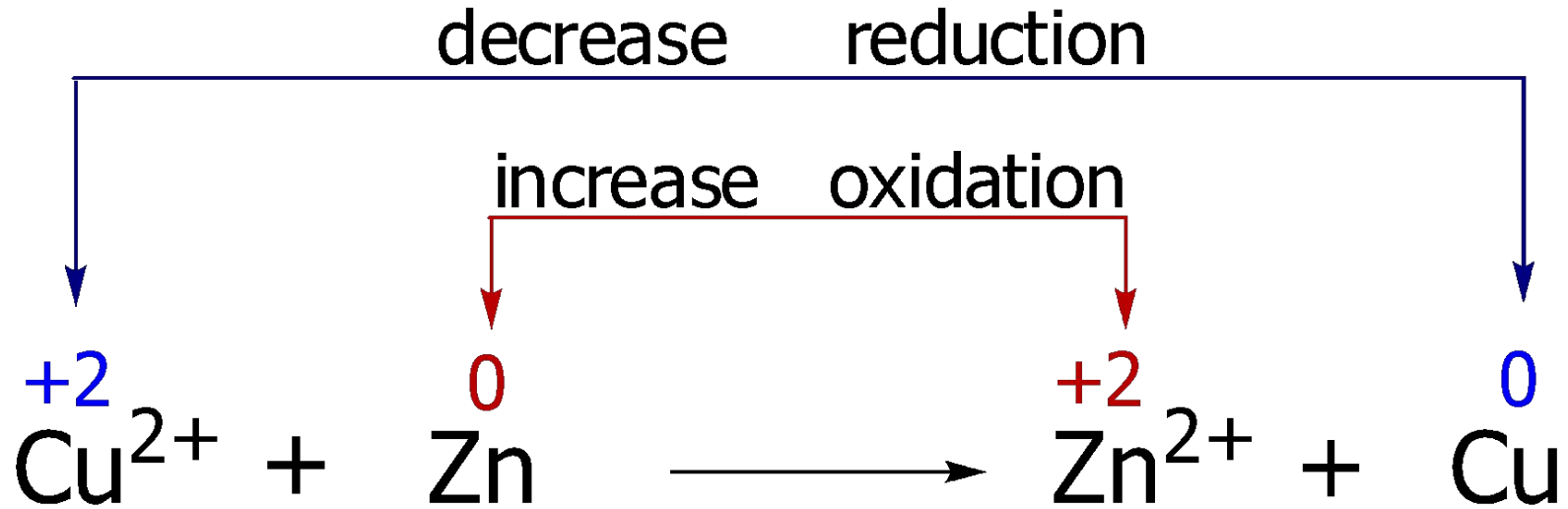
- **Increase in oxidation number**
- e^- loss

Reduction

- **Decrease in oxidation number**
- e^- gain

Using Oxidation Numbers to Recognize Redox Reactions

- Sometimes literal electron transfer:

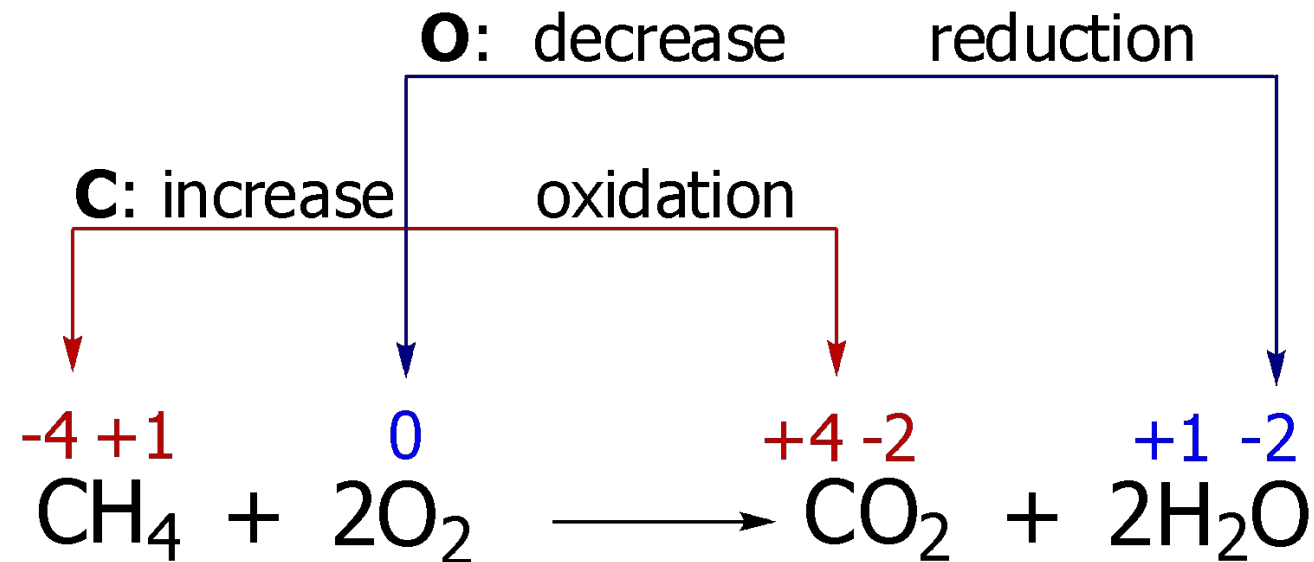


Cu: oxidation number decreases by 2
 $\Leftrightarrow$ reduction

Zn: oxidation number increases by 2
 $\Leftrightarrow$ oxidation

Using Oxidation Numbers to Recognize Redox Reactions

- Sometimes electron transferred in "formal" sense.



- **O:** oxidation number decreases by 2
 $\Leftrightarrow$ reduction
- **C:** oxidation number increases by 8
 $\Leftrightarrow$ oxidation

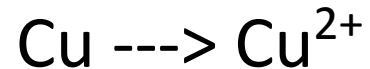
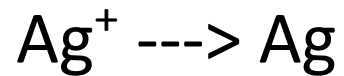
What is the oxidation number of



What is a Half-Reaction?

- A half-reaction is simply one which shows either reduction OR oxidation, but not both. Here is an example redox reaction:
- $\text{Ag}^+ + \text{Cu} \rightarrow \text{Ag} + \text{Cu}^{2+}$
- It has BOTH a reduction and an oxidation in it. That is why we call it a redox reaction, from **RED**uction and **OX**idation.

- What you must be able to do is look at a redox reaction and separate out the two half-reactions in it. To do that, identify the atoms which get reduced and get oxidized. Here are the two half-reactions from the example:



- The silver is being reduced, its oxidation number going from +1 to zero. The copper's oxidation number went from zero to +2, so it was oxidized in the reaction. In order to figure out the half-reactions, you MUST be able to calculate the oxidation number of an atom.

- When you look at the two half-reactions, you will see they are already balanced for atoms with one Ag on each side and one Cu on each side. So, all we need to do is balance the charge.
- To do this you add electrons to the more positive side. You add enough to make the total charge on each side become EQUAL.
- To the silver half-reaction, we add one electron:



- To the copper half-reaction, we add two electrons:

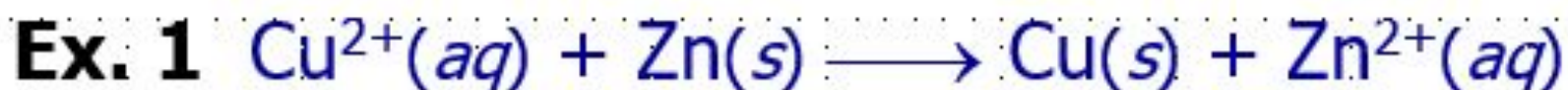


Half-reactions NEVER occur alone.

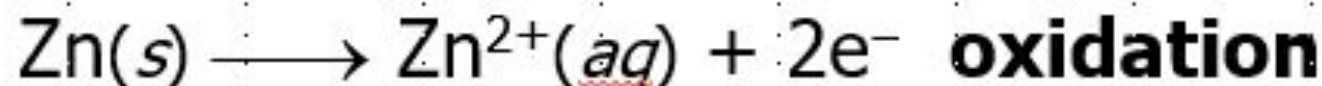
notice that each half-reaction wound up with a total charge of zero on each side. This is not always the case. You need to strive to get the total charge on each side EQUAL, not zero.

Balancing Redox Reactions

Some Redox reactions are simple:

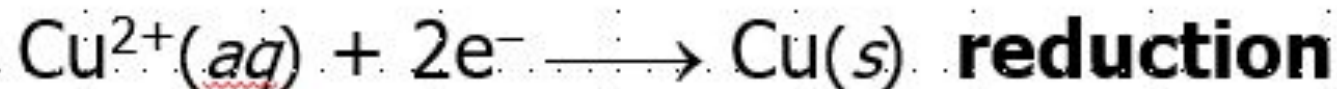


Break into half-reactions



Reducing agent

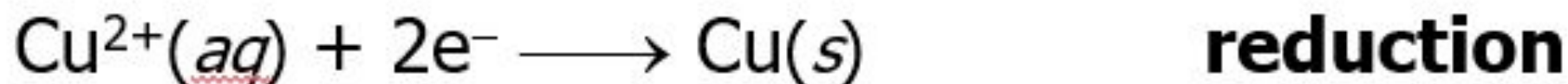
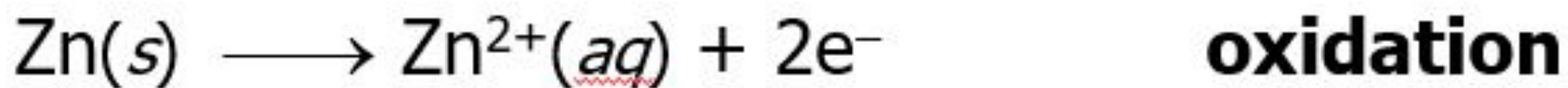
LEO



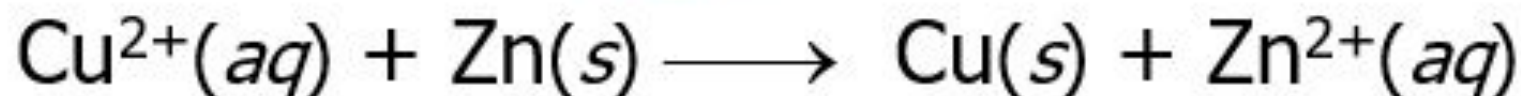
Oxidizing agent

GER

Example 1



- Each half-reaction is balanced for atoms
 - Same # atoms of each type on each side
- Each half-reaction is balanced for charge
 - Same sum of charges on each side
- Add both equations algebraically, canceling e^{-} 's
- **NEVER have e^{-} 's in net ionic equation**



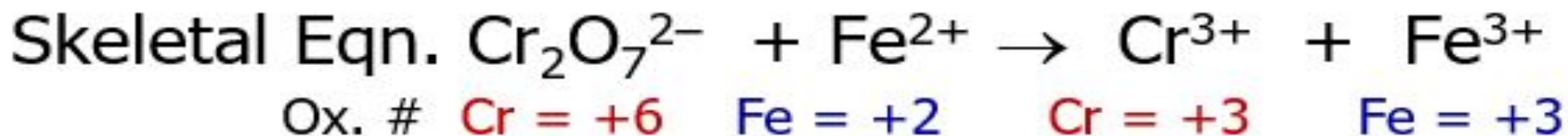
Balancing Redox Equations in Aqueous Solutions

- Many redox reactions in aqueous solution involve H_2O and H^+ or OH^-
- Balancing the equation cannot be done by inspection.
- Need method to balance equation correctly
- Start with **acidic** solution then work to basic conditions

Redox in Aqueous Solution

Ex. 2 Mix solutions of $\text{K}_2\text{Cr}_2\text{O}_7$ & FeSO_4

- Dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$, oxidizes Fe^{2+} to Fe^{3+}
- $\text{Cr}_2\text{O}_7^{2-}$ is reduced to form Cr^{3+}
- Acidity of mixture decreases as H^+ reacts with oxygen to form water



Ion-Electron Method – Acidic Solution

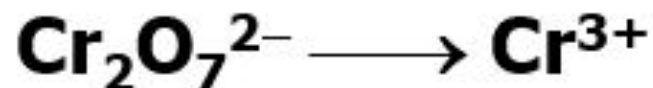
1. Divide equation into 2 half-reactions
2. Balance atoms other than H & O
3. Balance O by adding H_2O to side that needs O
4. Balance H by adding H^+ to side that needs H
5. Balance net charge by adding e^-
6. Make e^- gain equal e^- loss; then add half-reactions
7. Cancel anything that is the same on both sides

Ion Electron Method

Ex. 2 Balance in Acidic Solution



1. Break into half-reactions



2. Balance atoms other than H & O



- Put in 2 coefficient to balance Cr



- Fe already balanced

Ex. 2 Ion-Electron Method in Acid

3. Balance O by adding H_2O to the side that needs O.



- Right side has 7 O atoms
- Left side has none
- Add 7 H_2O to left side



- No O to balance

Ex. 2 Ion-Electron Method in Acid

4. Balance H by adding H^+ to side that needs H



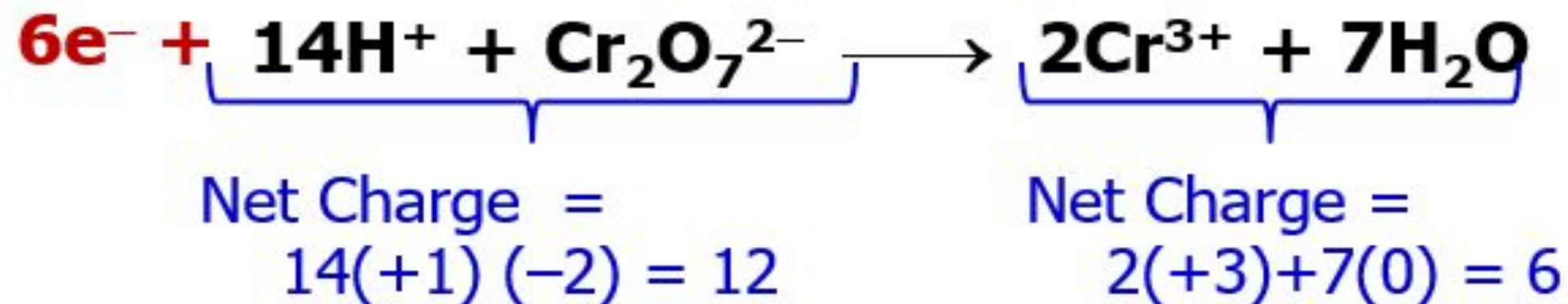
- Left side has 14 H atoms
- Right side has none
- Add 14 H^+ to right side



- No H to balance

Ex. 2 Ion-Electron Method in Acid

5. Balance net charge by adding electrons.



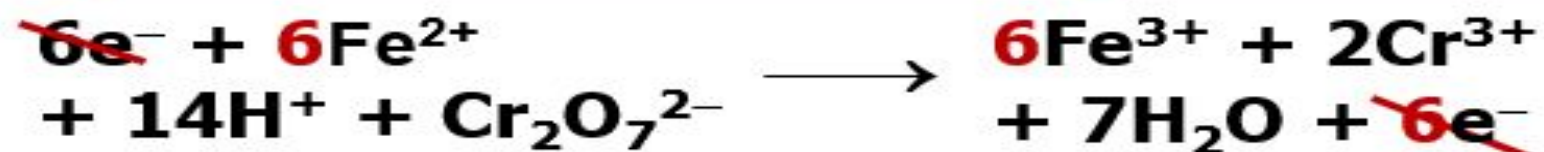
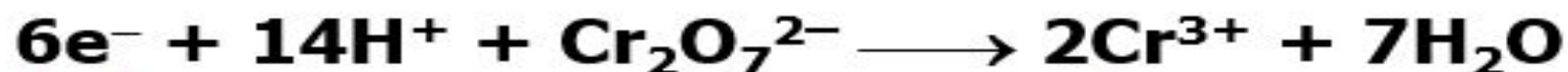
- 6 electrons must be added to reactant side



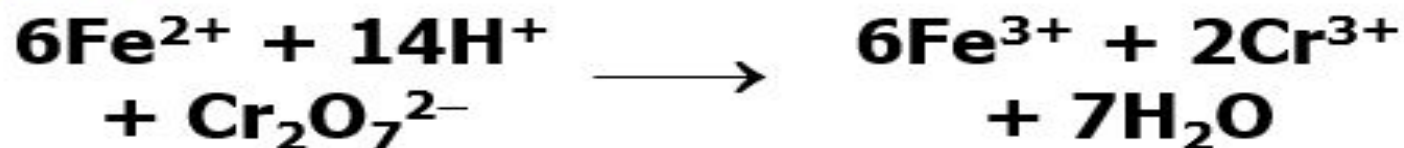
- 1 electron must be added to product side
- Now both half-reactions balanced for mass & charge

Ex. 2 Ion-Electron Method in Acid

6. Make e^- gain equal e^- loss; then add half-reactions



7. Cancel anything that's the same on both sides



Ion-Electron in Basic Solution

- The simplest way to balance an equation in **basic solution**

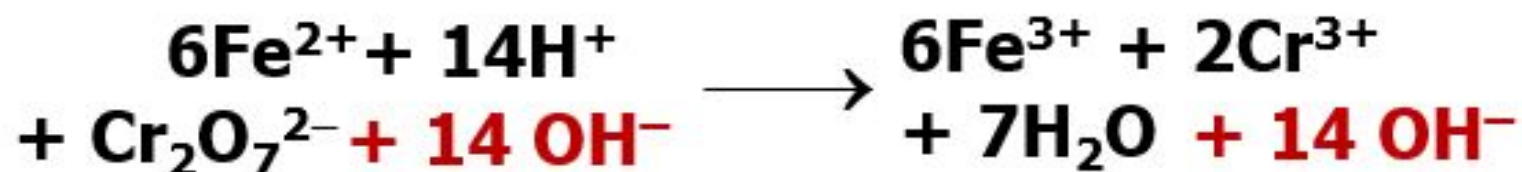
Use steps 1-7 above, then

8. Add the same number of OH^- to ***both*** sides of the equation as there are H^+ .
9. Combine H^+ & OH^- to form H_2O
10. Cancel any H_2O that you can from both sides

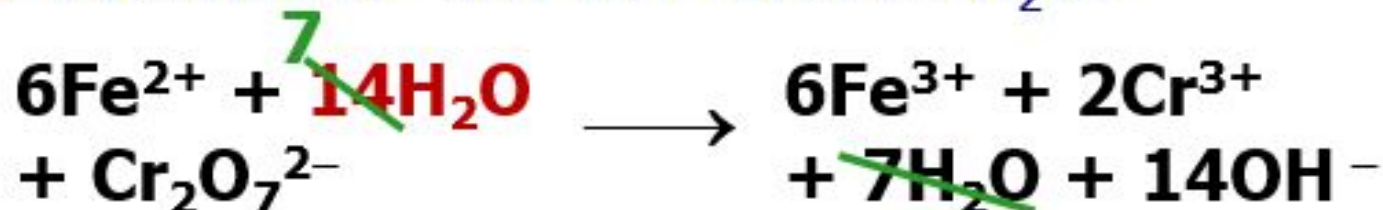
Ex.2 Ion-Electron Method in Base

Returning to our example of $\text{Cr}_2\text{O}_7^{2-}$ & Fe^{2+}

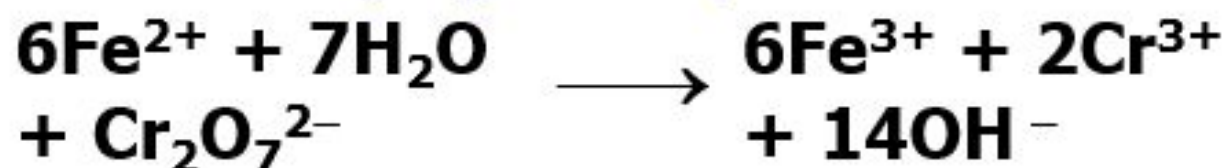
8. Add to **both** sides of equation the same number of OH^- as there are H^+ .



9. Combine H^+ and OH^- to form H_2O .

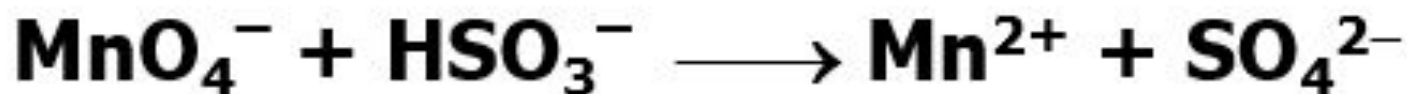


10. Cancel any H_2O that you can



Ex. 3 Ion-Electron Method

Balance the following equation in basic solution:



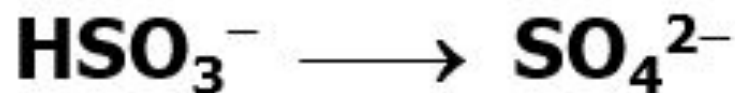
1. Break it into half-reactions



2. Balance atoms other than H & O



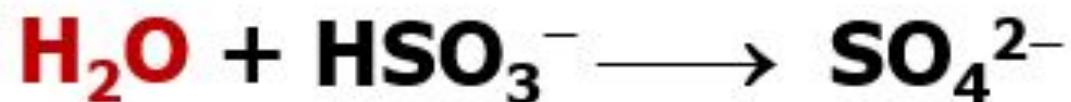
- Balanced for Mn



- Balanced for S

Ex. 3 Ion-Electron Method

3. Add H_2O to balance O

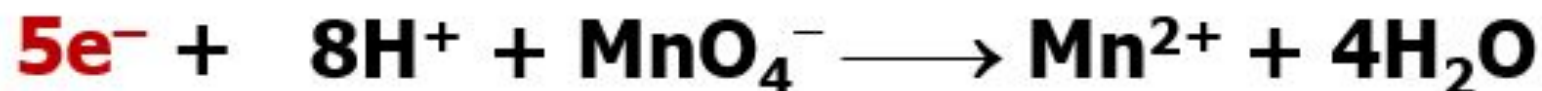


4. Add H^+ to balance H



Ex. 3 Ion-Electron Method

5. Balance net charge by adding e^- .



$$8 \times (+1) + (-1) = +7 \qquad +2 + 0 = +2$$

Add 5 e^- to reactant side

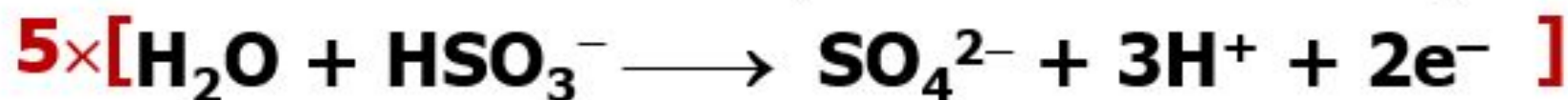
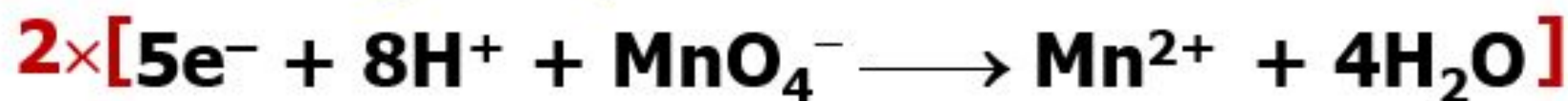


$$0 + (-1) = -1 \qquad -2 + 3(+1) = +1$$

Add 2 e^- to product side

Ex. 3 Ion-Electron Method

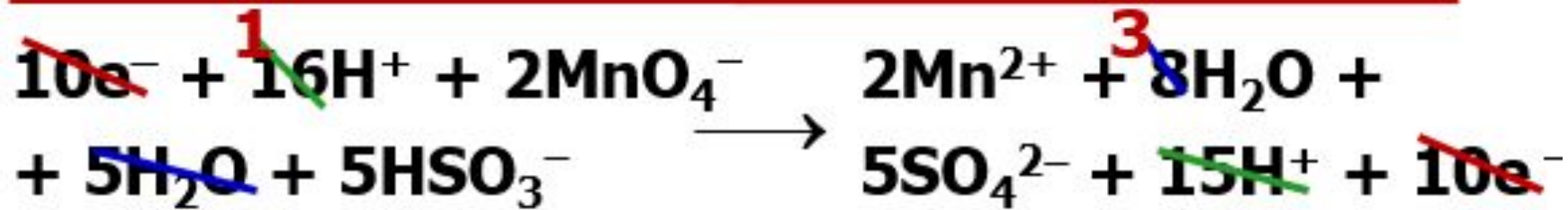
6. Make e^- gain equal e^- loss



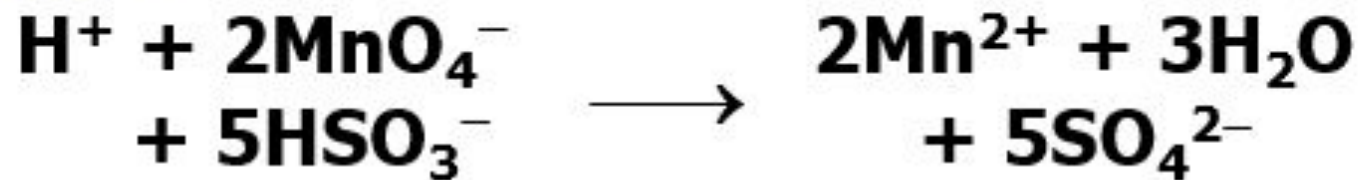
- Must multiply Mn half-reaction by 2
- Must multiply S half-reaction by 5
- Now have 10 e^- on each side

Ex. 3 Ion-Electron Method

6. Then add the two half-reactions



7. Cancel anything that is the same on both sides.



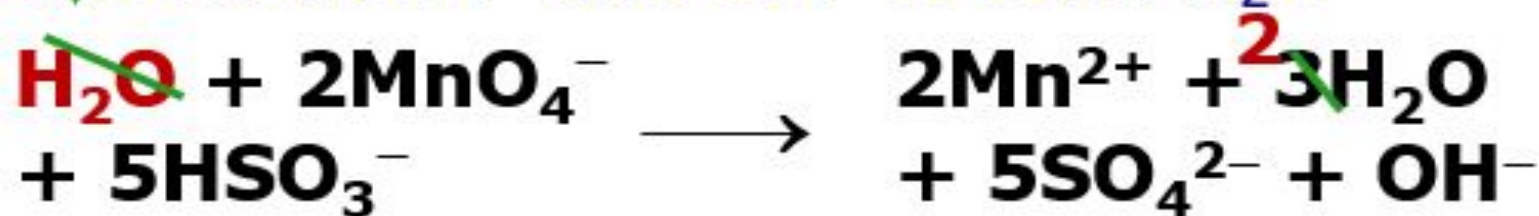
Balanced in acid.

Ex.3 Ion-Electron Method in Base

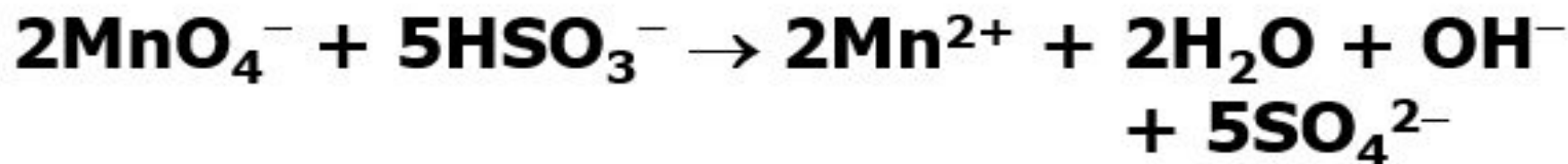
8. Add same number of OH^- to **both** sides of equation as there are H^+



9. Combine H^+ and OH^- to form H_2O

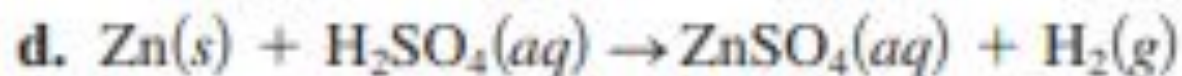
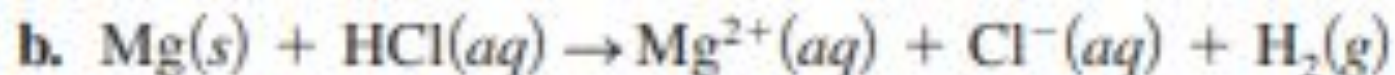
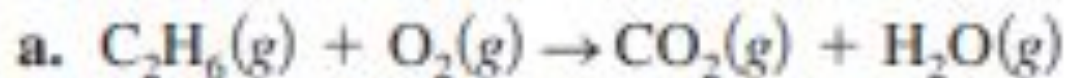


10. Cancel any H_2O that you can



Problems

Balance each of the following oxidation–reduction reactions by using the oxidation states method.



Balance each of the following oxidation–reduction reactions by using the oxidation states method.

